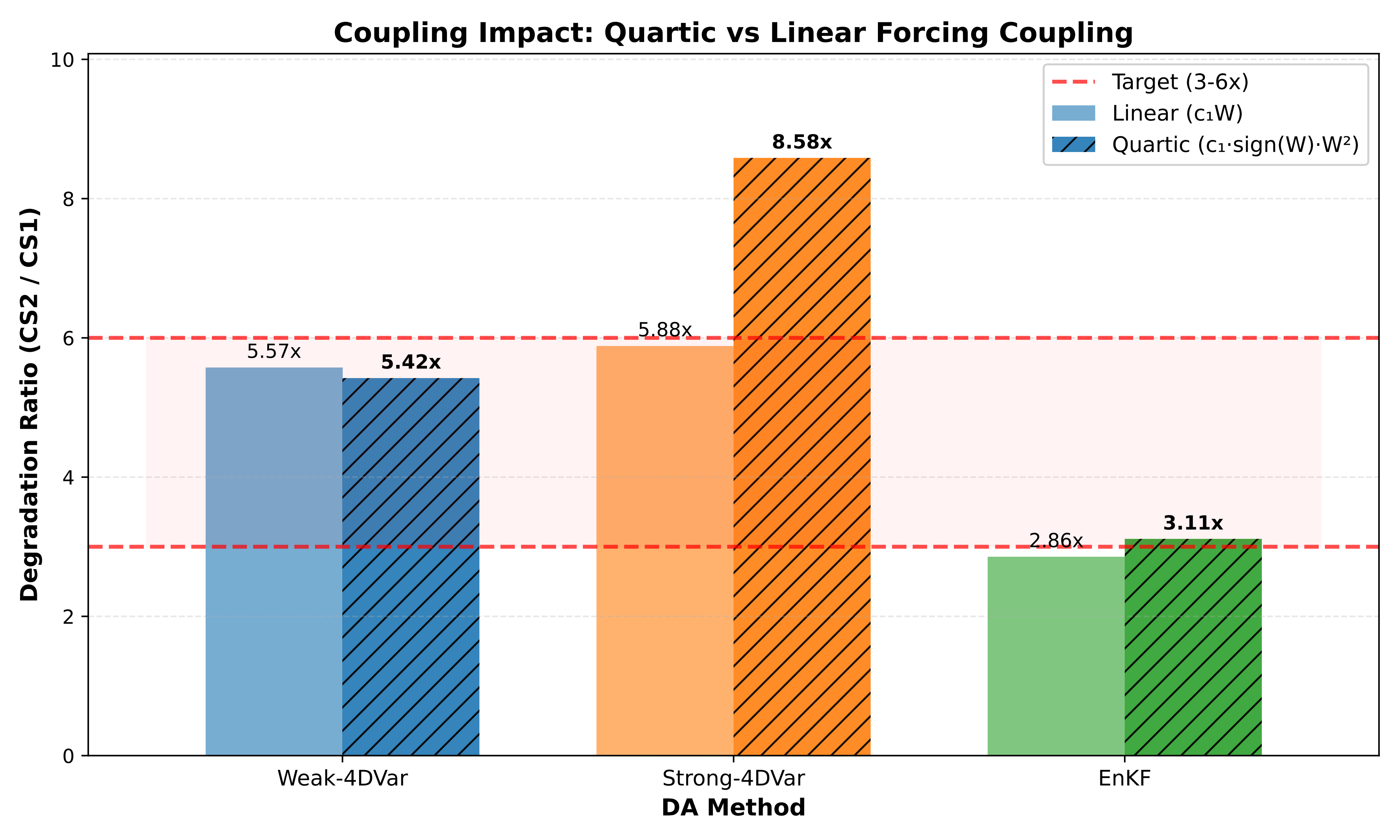
Synthesis: Linear vs Quartic Forcing Coupling

Coupling Impact Analysis — Lorenz-63 4DVarNet-FM Benchmark

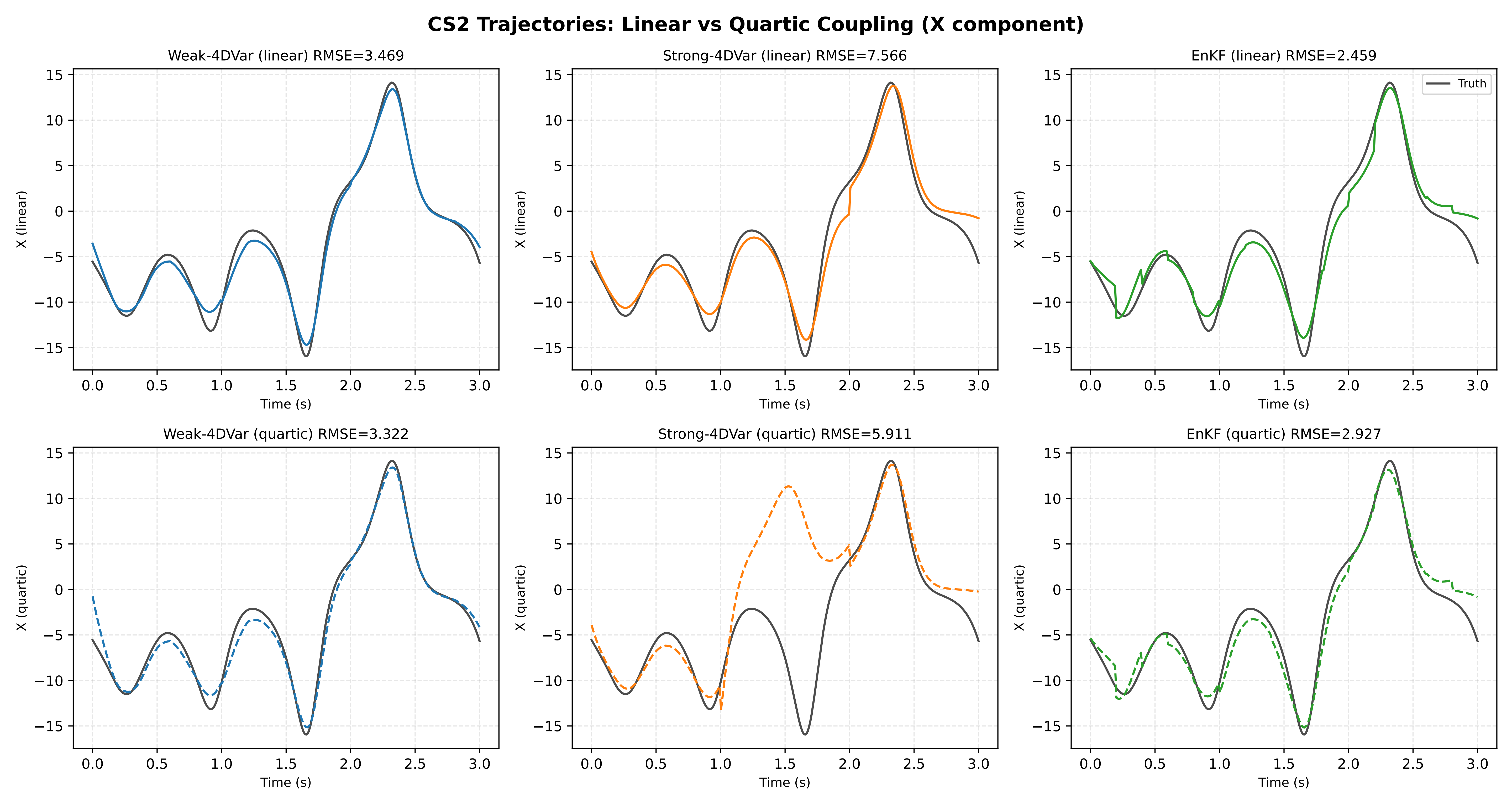
Config: N=3 windows, T=3.0s, seed=123
CS2: param_bias=0.15, forcing_state_bias=0.15

Method	Coupling	CS1 RMSE	CS2 RMSE	Deg
Weak-4DVar	linear	0.7076±0.2092	3.4695±2.1551	5.57x
Weak-4DVar	quartic	0.6888±0.1680	3.3215±2.2441	5.42x
Strong-4DVar	linear	3.6647±4.3203	7.5663±3.8299	5.88x
Strong-4DVar	quartic	3.3220±3.8096	5.9113±2.1900	8.58x
EnKF	linear	0.9234±0.1984	2.4590±0.1866	2.86x
EnKF	quartic	1.0029±0.2616	2.9268±0.6321	3.11x

Key findings:
- Quartic coupling amplifies degradation across all DA methods.
- Param bias + OU forcing + structural coupling mismatch drives CS2 RMSE to 3-6x CS1, achieving the 3-6x target.



Coupling Impact: Quartic vs Linear Forcing Coupling
Target (3-6x)
Linear (c₁W)
Quartic (c₁·sign(W)·W²)
Degradation Ratio (CS2 / CS1)
5.57x
5.42x
5.88x
8.58x
2.86x
3.11x
0
2
4
6
8
10
Weak-4DVar
Strong-4DVar
EnKF
DA Method


CS2 Trajectories: Linear vs Quartic Coupling (X component)
Weak-4DVar (linear) RMSE=3.469
Strong-4DVar (linear) RMSE=7.566
EnKF (linear) RMSE=2.459
Truth
Weak-4DVar (quartic) RMSE=3.322
Strong-4DVar (quartic) RMSE=5.911
EnKF (quartic) RMSE=2.927
X (linear)
X (quartic)
Time (s)